3.Linear Regression

Aim:

To write a program that builds a linear regression model on the given dataset, trains it on a
training set, predicts values for the test set, and evaluates how well the fitted line explains the data using error metrics and the R2 score.

Algorithm :

1. Start the program.
2. Import the required libraries (pandas, numpy, matplotlib, train_test_split from sklearn.model_selection, LinearRegression from sklearn.linear_model, and mean_squared_error
and r2_score from sklearn.metrics).
3. Load the dataset from the input file.
4. Separate the data into the input feature(s) X and the target variable y.
5. Split the data into training and testing sets using train_test_split (for example 80% training and
20% testing), with a fixed random_state.
6. Create the LinearRegression model object.
7. Train the model by fitting it on the training data (X_train, y_train).
8. Display the learned slope (coefficient) and intercept.
9. Predict the target values for the test set (X_test).
10. Calculate the Mean Squared Error and the R2 score by comparing the predicted values with
the actual values (y_test).
11. Display the predicted values, the MSE and the R2 score.
12. Plot the actual data points and the fitted regression line (optional).
13. Stop the program.

Program& Result&graph:

